

A. Design finite automata for following Languages over alphabet { a, b}

1. Number of a's is multiple of 3
2. Number of a's are odd and b's are odd
3. String of length multiple of 5
4. String length atmost 5
5. All strings with exactly two a's
6. All strings containing at most two b's
7. All strings starting with aaa
8. All strings with aaa as substring
9. All strings ending in aaa

B. Design FA for string of {0,1} binary number divisible by 4

C. Convert the NFA to DFA

1.

	0	1
$\rightarrow q_0$	ϕ	q_1
q_1	q_2	q_1
$*q_2$	ϕ	q_1

2.

	0	1
$\rightarrow p$	{p, q}	{p}
q	{r, s}	{t}
r	{p, r}	{t}
*s	ϕ	ϕ
*t	ϕ	ϕ